CERTIFICATE OF VERIFICATION

Note: This table completed by ~~HERS~~ Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Design ~~HERS~~ Verified Dwelling Unit Water Heating Systems Information (other than HPWH)

This table reports features of the water heating system(s) ~~-~~other than HPWH system specified on the registered LMCC compliance document for this project.

01	02	03	04	05	06	07	08	09	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Central DHW System Distribution	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery

A2. Design ~~HERS~~Verified Dwelling Unit HPWH System Information

This table reports the water heating system(s) that were specified on the registered LMCC compliance document for this project.

01	02	03	04	05	06	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Dwelling Unit DHW System Distribution Type	Compact Distribution	Drain Water Heat Recovery	Simulated Equipment Make and Model

B. Installed ~~HERS~~Verified Dwelling Unit Water Heating Systems Information

This table reports the water heating system features installed in this project.

01	02	03	04	05	06	07	08	09	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Central DHW System Distribution	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery

B2. Installed ~~HERS~~ Verified Dwelling Unit HPWH System Information

This table reports the water heating system(s) installed in this project.

01	02	03	04	05	06	07	08	09
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Dwelling Unit DHW System Distribution Type	Compact Distribution	Drain Water Heat Recovery

C. Design ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heater(s) efficiency features specified on the registered LMCC compliance document for this project. (Not needed for central systems)

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank location

D. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heater(s) efficiency features installed in this project. (Not needed for central systems)

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank location
08	Compliance Statement					

E. Installed Water Heater Manufacturer Information

01	02	03
Water Heating System ID or Name	Manufacturer	Model Number

F. Mandatory Measures for all Domestic Hot Water Distribution Systems

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

01	Equipment shall meet the applicable requirements of the Appliance Efficiency Regulations (Section 110.3(b)1).
02	Unfired storage tanks are insulated with an external R-3.5 or combination of R-16 internal and external Insulation. (Section 110.3(c)3).
03	Domestic hot water piping insulation requirements (Section 150(J)): • All domestic hot water piping shall be insulated as specified in Section 609.12 of the California Plumbing Code. Insulation buried below grade must be installed in a waterproof and non-crushable casing or sleeve. • Pipe insulation shall fit tightly and all elbows and tees shall be fully insulated. • Piping that penetrates framing members shall not be required to have pipe insulation for the distance of the framing penetration. • Piping that penetrates metal framing shall use grommets, plugs, wrapping or other insulating material to assure that no contact is made with the metal framing. Insulation shall butt securely against all framing members. • Piping surrounded with a minimum of 1 inch of wall insulation, 2 inches of crawlspace insulation, or 4 inches of attic insulation shall not be required to have pipe insulation. • Insulation is not required on the cold water line when it is used as the return.
04	For Gas or Propane Water Heaters: Ensure either a **or** b are installed (Section 150.0(n)) a) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall **within 3 feet** from the water heater • A dedicated 125V, 20A electrical receptacle connected to the electric panel with a 120/240V 3 conductor, branch circuit rated at 30 amps minimum, within 3 feet from the water heater and is accessible with no obstructions. • The conductor shall be labeled with the word "Spare" on both ends; and • A reserved single pole circuit breaker space next to the circuit breaker next to the branch circuit labeled "Future 240V use" shall be provided. • A condensate drain no more than 2 inches higher than the base of the water heater, and allows for natural draining without pump assistance. b) A designated space at least 2.5 feet by 2.5 feet and 7 feet tall **more than 3 feet** from the water heater • A dedicated 240 volt branch circuit shall be installed within 3 feet from the designated space. The branch circuit shall be rated at 30 amps minimum. The blank cover shall be identified as "240V ready"; and • The main electrical service panel shall have a reserved space to allow for the installation of a double pole circuit breaker for a future HPWH installation. The reserved space shall be permanently marked as "For Future 240V use"; and • Either a dedicated cold water supply, or the cold water supply shall pass through the designated HPWH location just before reaching the gas or propane water heater; and • The hot water supply pipe coming out of the gas or propane water heater shall be routed first through the designated HPWH location before serving any fixtures; and • The hot and cold water piping at the designated HPWH location shall be exposed and readily accessible for future installation of a HPWH; and • A condensate drain no more than 2 inches higher than the base of the installed water heater, and allows natural draining without pump assistance.
05	**Domestic hot water piping insulation requirements**: See the exceptions to Section 160.4(e) All piping for multifamily domestic hot water systems shall be insulated and meet the applicable requirements below: a. **General Requirements:** a. The first 8 feet of inlet cold water piping from the storage tanks, including piping between a storage tank and a heat trap shall be insulated. b. Insulation on the piping and domestic hot water system appurtenances shall be continuous. c. Pipe supports, hangers, and pipe clamps shall be attached on the outside of rigid pipe insulation to prevent thermal bridges. d. All pipe insulation seams shall be sealed. e. Insulation for pipe elbows shall be mitered, preformed, or site fabricated with PVC covers. f. Insulation for tees shall be notched, preformed, or site fabricated with PVC covers. g. Extended stem isolation valves shall be installed. h. All plumbing appurtenances on hot water piping from a heating source to heating plant, at the heating plant, and distribution supply and return piping shall be insulated to meet the following requirements: 1. Where the outer diameter of the appurtenance is less than the outer diameter of the insulated pipe that it is attached to, the appurtenance shall be insulated flush with the insulation surrounding the pipe. 2. Where the outer diameter of the appurtenance is greater than the outer diameter of the insulated pipe that it is attached to, the appurtenance shall be insulated with a minimum thickness of 1 inch.

	3. The insulation shall be removable and re-installable to ensure maintenance or replacement services can be completed. 4. Valves shall be fully functional without impediment from the insulation. b. **Insulation Thickness:** All piping for multifamily domestic hot water systems shall meet the insulation thickness requirements specified in of Table 160.4-A. a. For insulation conductivity in the range shown in Table 160.4-A for the applicable fluid temperature range, the insulation shall have the applicable minimum thickness or R-value shown in Table 160.4-A. b. if the insulation conductivity falls outside the range provided in Table 160.4-A applicable fluid temperature range, the insulation shall meet a minimum R-value as indicated in Table 160.4-A. Or, it can have a thickness determined using Equation 160.4-A. c. Insulation conductivity shall be determined in accordance with ASTM C335 at the mean temperature listed in Table 160.4-A, and shall be rounded to the nearest 1/100 Btu-inch per hour per square foot per °F. c. **Insulation Protection:** Pipe Insulation shall be protected from damage due to sunlight, moisture, equipment maintenance and wind. Protection shall, at minimum, include the following: a. Pipe and appurtenance insulation exposed to weather shall be protected by a cover suitable for outdoor service. The cover shall be water retardant and provide shielding from solar radiation that can cause degradation of the material. Appurtenance insulation covers shall be removable and able to be reinstalled. Adhesive tape shall not be used to provide this protection. b. Pipe insulation covering chilled water piping and refrigerant suction piping located outside the conditioned space shall include, or be protected by, a Class I or Class II vapor retarder. All penetrations and joints shall be sealed. c. Pipe insulation buried below grade must be installed in a waterproof and noncrushable casing or sleeve.	
06~~5~~	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
0~~6~~7	Correction Notes:	

G. ~~HERS~~ Verified Compact Hot Water Distribution System Expanded Credit (CHWDS-H-EX) (RA3.6.5)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor

10	No hot water piping >1 inch diameter is allowed.	
11	Length of 1 inch diameter piping is limited to 8 feet or less.	
12	Two and three story buildings cannot have hot water distribution piping in the attic, unless the water heater is also located in the attic.	
13	Eligible recirculating systems must be ~~HERS~~ Verified Demand Recirculation: Manual Control conforming to RA4.4.17.	
14	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
15	Correction Notes:	

H. Compact Hot Water Distribution System ~~(CHWDS)~~ (RA4.4.6)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

For dwelling units with multiple systems, enter the master bath distance and kitchen distance to the closest water heater, and enter the average of the furthest fixture to each water heater.

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor

I. ~~HERS~~ Verified Drain Water Heat Recovery System (DWHR-H) (RA3.6.9)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

DWHR devices shall comply with these requirements.

Design DWHR System Information			
01	02	03	04
System ID/Name	Rated Effectiveness	Installation Configuration	Percent of shower served by the DWHR device

Installed DWHR System Information						
05	06	07	08	09	10	11
System Name/ID	Manufacturer	Model #	Rated effectiveness	Installation Configuration	Percent of shower served by the DWHR device	DWHR System Certified by CEC (Yes/No)

12	For water heating system serving a single dwelling, the DWHR system shall, at the minimum, recover heat from the master bathroom shower and must at least transfer that heat either back to the respective shower(s) or the water heater.	
13	For central water heating system serving multiple dwellings, the DWHR system shall, at the minimum, recover heat from half the showers located above the first floor and must at least transfer that heat either back to all the respective showers or the water heater.	
14	The DWHR unit(s) shall be installed within 1 degrees of the rated slope. Sloped DWHR shall have a minimum lengthwise slope of 1 degree. The lateral level tolerance shall be within plus or minus 1 degree.	
15	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
16	Correction Notes:	

J. ~~HERS~~ Verified Pipe Insulation for Central Systems~~Credit~~ Requirements ~~(PIC-H)~~ (RA3.6.~~3~~2)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

Systems that utilize this distribution type shall comply with these requirements.

01	~~HERS~~ECC rater shall inspect the heating plant and horizontal supply header and return piping in accordance with the requirements in Title 24 Part 6 section 170.2(d).~~shall perform a visual inspection that all hot water piping comply with the insulation requirements in 150.0(J).~~	
02	The rater shall use a sampling approach that one in seven DHW recirculation pipe risers and associated branches be inspected to verify the pipe insulation meet with the following requirements: a. All piping for multifamily domestic hot water systems shall be insulated to the thickness specified in Table 160.4-A, including the first 8 feet of inlet cold water piping to the heating plant. Insulation on the piping and appurtenances shall be continuous. b. All appurtenances at the heating plant, from a heating source to storage tank(s), or in between storage tanks and storage water heaters, and recirculation supply and return loop shall meet the following: 1. Insulation to be flush with pipe insulation or have minimum of one inch if appurtenance is bulkier. 2. Removable and re-installable for maintenance or replacement. 3. Pipe supports, hangers, and clamps shall be attached on the outside of rigid pipe insulation. c. All pipe insulation seams shall be sealed along the length of the pipe and between adjacent sections of insulation material. d. Insulation for pipe elbows shall be mitered, and insulation for tees shall be notched. Alternatively, tees and elbows may be pre-formed, or site fabricated with PVC covers. Isolation valves shall be fully functional. Extended stem isolation valves shall be installed on hot water piping or where pipe insulation is required.	
03~~2~~	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
04~~3~~	Correction Notes:	

K. ~~HERS~~ Verified Central Parallel Piping Requirements (PP-H) (RA3.6.4)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

Systems that utilize this distribution type shall comply with these requirements.

01	Each central manifold has 5 feet or less of pipe between manifold and water heater.	
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code.	
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For example, piping from a second story manifold cannot supply the first floor.	
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A.	
05	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
06	Correction Notes:	

L. Central Parallel Piping Requirements ~~(PP)~~ (RA4.4.4)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Systems that utilize this distribution type shall comply with these requirements.

01	Each central manifold has 15 feet or less of pipe between manifold and water heater
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code.
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For instance, piping from a second story manifold cannot supply the first floor
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A-1.

M. Point of Use Requirements (POU) (RA4.4.5)

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Systems that utilize this distribution type shall comply with these requirements.

01	All hot water supply pipe run lengths are equal to or less than the maximum values shown below, based on the pipe diameter. If a combination of piping is used in a single run, then one half the allowed length of each size is the maximum installed length. The maximum allowed length of piping for the longest run terminating in: 3/8 inch - For only one pipe size - max length allowed is 15 feet For combination pipe sizes the max allowed length of 3/8-inch piping is 7.5 feet, of ½ inch piping is 5 feet, and ¾ inch piping is 2.5 feet. ½ inch - For only one pipe size – max length allowed is 10 feet For combination pipe sizes the allowed length of ½-inch piping is 5 feet, and ¾ inch piping is 2.5 feet. ¾ inch - For only one pipe size = 5 feet

N. Mandatory Requirements for all Recirculation Systems (RA4.4.7)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met

01	A check valve located between the recirculation pump and the water heater to prevent unintentional recirculation.
02	Piping must take most direct path between water heater and fixtures.
03	Insulation is not required on the cold water line when it is used as the return.
~~04~~	~~If more than one loop installed each loop shall have its own pump and controls.~~

O. Recirculation Non-Demand Controls Requirements ~~(R-ND)~~ (RA4.4.8)

Systems that utilize this distribution type shall comply with these requirements.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The active control shall be either: timer, temperature, or time and temperature. Timers shall be set to less than 24 hours. The temperature sensor shall be connected to the piping and to the controls for the pump.

P. Demand Recirculation Manual Control ~~(R-DRmc)~~ (RA4.4.9)/Sensor Control ~~(RDRsc)~~ (RA4.4.10) Requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Systems that utilize this distribution type shall comply with these requirements.

01	The system operates "on-demand", meaning that the pump begins to operate shortly before or immediately after hot water draw begins, and stops when the return water temperature reaches a certain threshold value. For Demand Recirculation Manual Control, the pump shall be turned on using a manual switch system. For Demand Recirculation Sensor Control, the pump shall be turned on using a sensor system.
02	The controls shall be located in the kitchen, bathroom, and any hot water fixture location that is at least 20 feet from the water heater.
03	Manual controls may be active by wired or wireless mechanisms. Each control shall have standby power of 1 Watt or less.
04	Sensor Controls may be activated by wired or wireless mechanisms, including buttons, motion sensors, door switches and flow switches. Each control shall have standby power of 1 Watt or less.
05	Pump and control placement shall meet one of the following criteria: • When a dedicated return line has been installed the pump, controls and thermo-sensor are installed at the end of the supply portion of the recirculation loop; or • The pump and controls are installed on the dedicated return line near the water heater and the thermo-sensor is installed in an accessible location as close to the end of the supply portion of the recirculation loop as possible; or • When the cold water line is used as the return, the pump, demand controls and thermo-sensor shall be installed in an accessible location at the end of supply portion of the hot water distribution line (typically under a sink).
06	After the pump has been activated, the controls shall allow the pump to operate until the water temperature at the thermo-sensor rises to one of the following values: • Not more than 10°F (5.6°C) above the initial temperature of the water in the pipe • Not more than 102°F (38.9°C).
07	Controls shall limit operation to no more than 5 minutes following activation.

Q. ~~HERS~~ Verified Demand Recirculation Manual Control (RDRmc-H) (RA3.6.6)/Sensor Control (RDRsc-H) (RA3.6.7) Requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

Systems that utilize this distribution type shall comply with these requirements.

01	~~HERS~~ECC rater shall perform a visual inspection to verify that the demand pump, manual/sensor controls and thermo-sensor are present and operating properly consistent with the applicable requirements of RA4.4.9 and RA4.4.10	
02	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
03	Correction Notes:	

R. Determination of ~~HERS~~ECC Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

01	

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Company:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Verification is true and correct.
2. I am the certified ~~HERS~~ECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater).
3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require ~~HERS~~ECC verification identified on this Certificate of Verification comply with the applicable requirements in Reference Appendices RA2, RA3, and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency.
4. The information reported on applicable sections of the Certificate(s) of Installation (LMCI) signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (LMCC) approved by the enforcement agency.
5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building.

BUILDER OR INSTALLER INFORMATION AS SHOWN ON THE CERTIFICATE OF INSTALLATION

Company Name (Installing Subcontractor, General Contractor, or Builder/Owner):	
Responsible Builder or Installer Name:	CSLB License:

~~HERS~~ECC PROVIDER DATA REGISTRY INFORMATION

Sample Group Number (if applicable):	Dwelling Test Status in Sample Group (if applicable):

~~HERS~~ECC RATER INFORMATION

~~HERS~~ECC Rater Company Name:	
Responsible Rater Name:	Responsible Rater Signature:
Responsible Rater Certification Number w/ this ~~HERS~~ECC Provider	Date Signed:

LMCV-PLB-22-H User Instructions

A. Design ~~HERS~~ Verified Central Water Heating Systems Information

This table reports the water heating system features that were specified on the registered LMCC compliance document for this project. For information only and requires no user input.

A1. Design ~~HERS~~ Verified Dwelling Unit HPWH System Information

This table reports the water heating system features that were specified on the registered LMCC compliance document for this project. This section is for information/verification purposes only and requires no user input.

B. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Systems Information

This table reports the water heating system information that is being installed. Require one line for each system.

1. Dwelling Unit Name - Reference information from LMCC.
2. Water Heating System ID or Name – Reference information from LMCC.
3. Water Heating System Type – Reference information from LMCC. The different kinds of water heating system type are DHW, or Combined Hydronic.
4. Water Heater Type – Information from LMCC. The different kinds of water heaters are Large/Commercial Storage, Small/Consumer Storage, Residential-Duty Commercial Storage, Heat Pump, Boiler, Large/Commercial Instantaneous, Small/Consumer Instantaneous, Residential-Duty Commercial Instantaneous or Indirect.
5. # of Like (or Identical) Water Heaters in system – Reference information from LMCC.
6. 06 Fuel Type – Reference information from LMCC. The different kinds of fuel types are natural gas, propane, oil, or electricity.
7. Rated Input Type – Reference information from LMCC. For natural gas, propane and oil fuel type the input type is Btu/hr. For electric the input type is kW.
8. Rated Input Value – User input. Numerical value of the rated input. Must be equal to or less than value indicated on the LMCC.
9. Central DHW System Distribution - Reference information from LMCC.
10. Dwelling Unit DHW System Distribution Type - Reference information from LMCC.
11. Compact Distribution - Reference information from LMCC.
12. Drain Water Heat Recovery - Reference information from LMCC.

B2. Installed ~~HERS~~ Verified Dwelling Unit HPWH System Information

This table reports the water heating system information that is being installed. Require one line for each installed water heater. Not applicable for central systems.

1. Dwelling Unit Name – Reference information from Table A2.
2. Water Heating System ID or Name – Reference information from Table A2. AFUE, UEF and Thermal Efficiency.
3. Modeled Equipment Make and Model – User input must be equal to the value indicated on Table C as default and allow user to override with an equivalent system based on the simulated equipment in Table A2.
4. # of Like (or Identical) Water Heaters in System – Reference information from Table A2

5. .Tank Location – User input. Must be equal to value indicated in Table B2.
6. Exterior Tank Insulation R-value – User input. Must be equal to or higher than value indicated in Table A2.
7. Dwelling Unit DHW System Distribution Type – Reference information from Table A2.
8. Compact Distribution — Reference information from Table A2.

C. Design ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heating system features that were specified on the registered LMCC compliance document for this project. For information only and requires no user input.

D. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

This table reports the water heating system information that is being installed. Require one line for each central system.

1. Water Heating System ID or Name – Reference information from LMCC
2. Heating Efficiency Type – Reference information from LMCC. Different efficiency types are Energy Factor, AFUE, UEF and Thermal Efficiency.
3. Heating Efficiency Value – User input. Numerical value of the Heating Efficiency. Must be equal to or higher efficiency than value indicated on the LMCC.
4. Standby Loss – User input. Must be equal to or less than value indicated on the LMCC. Value may be N/A if LMCC value is N/A.
5. Exterior Insulation R-Value – User input. Must be equal to or higher than value indicated on the LMCC. Value may be N/A if LMCC value is N/A.
6. Water Heater Storage Volume (gal) – User input. Value may be N/A if water heater type is instantaneous with zero storage.
7. Tank location – User input. Must be equal to system type indicated on the LMCC.

E. Installed Water Heater Manufacturer Information

This table reports the manufacturer information of the installed water heater(s). Require one line for each installed water heater

1. Water Heating System ID or Name – Reference information from LMCC.
2. Manufacturer – User input. Enter the name of the water heater manufacturer.
3. Model Number – User input. Enter the model number of the water heater.

F. Mandatory Measures for all Domestic Hot Water Distribution Systems

This table lists the requirements for all DHW systems. ~~HERS~~ECC rater must ensure all the requirements in this table are met.

G. ~~HERS~~ Verified Compact Hot Water Distribution Expanded System Credit and
H. Compact Hot Water Distribution System Basic

If performance compliance is used, this table lists the values used in the performance calculation and require no user input.

If prescriptive compliance is used, fill out this table

1. Reference information from LMCC
2. User select – user select from list

3. Enter the Master Bath distance of furthest fixture to Water Heater in feet. For multiple water heaters, enter the distance to the closest water heater.
4. Enter the Kitchen distance from furthest fixture to Water Heater in feet. For multiple water heaters, enter the distance to the closest water heater.
5. Enter Furthest Third fixtures from fixture to Water Heater in feet. For multiple water heaters, enter the average of the furthest distance of each water heater.
6. Calculated value – no user input required
7. Calculated value – no user input required

I. ~~HERS-~~Verified Drain Water Heat Recovery System

This table lists the requirements for all central recirculation systems. ~~HERS~~ECC rater must ensure all the requirements in this table are met.

1. Reference information from LMCC.
2. Reference information from LMCC.
3. Reference information from LMCC.
4. Reference information from LMCC.
5. Reference information from LMCC.
6. Drain Water Heat Recovery Manufacturer's name- Enter the name of the Manufacturer.
7. Drain Water Heat Recovery Manufacturer's model number – Enter the Model number.
8. Rated Effectiveness' – Enter the rated effectiveness of the DWHR device.
9. Installation Configuration – Enter type of configuration. Available options are Equal flow, unequal to shower, and unequal to water heater
10. Percent of shower served by the DWHR device – Enter the percent of showers served by this DWHR device.
11. DWHR System Certified by CEC – Enter "Yes" if certified or else enter "No".

J. ~~HERS-~~Verified Pipe Insulation for Central Systems ~~Credit~~ Requirements

This table only applies to systems indicated as **~~HERS~~-Verified Pipe Insulation for Central Systems ~~Credit~~.** In addition to the mandatory requirements in Table F, the ~~HERS~~ECC rater must ensure the requirements in this table are met.

K. ~~HERS-~~Verified Central Parallel Piping Requirements

This table only applies to systems indicated as **~~HERS~~-Verified Central Parallel Piping.** In addition to the mandatory requirements in Table F, the ~~HERS~~ rater must ensure the requirements in this table are met.

L. Central Parallel Piping Requirements

This table only applies to systems indicated as Central **Parallel Piping.** In addition to the mandatory requirements in Table F, the installer must ensure the requirements in this table are met.

M. Point of Use Requirements

This table only applies to systems indicated as **Point of Use** In addition to the mandatory requirements in Table F, the installer must ensure the requirements in this table are met.

N. Mandatory Requirements for all Recirculation Systems

The requirements of this table apply to all recirculation systems listed below.

O. Recirculation Non-Demand Controls Requirements
This table only applies to systems indicated as **Recirculation Non-demand controls.** In addition to the mandatory requirements in Table F and N, the installer must ensure the requirements in this table are met.

P. Demand Recirculation Manual Control/Sensor Control Requirements
This table only applies to systems indicated as **Demand Recirculation Manual Control, Demand Recirculation Senor Control, ~~HERS~~-Verified Demand Recirculation Manual Control** or **~~HERS~~-Verified Demand Recirculation Senor Control.** In addition to the mandatory requirements in Table F and N, the installer must ensure the requirements in this table are met.

Q. ~~HERS-~~Verified Demand Recirculation Manual Control (RDRmc-H) (RA3.6.6)/Sensor Control (RDRsc-H) (RA3.6.7)
This table only applies to systems indicated as **~~HERS~~-Verified Demand Recirculation Manual Control** or **~~HERS~~-Verified Demand Recirculation Senor Control.** In addition to the mandatory requirements in Table F and N, the ~~HERS~~ rater must ensure the requirements in this table are met.

R. Determination of ~~HERS~~ Verification Compliance
This field is filled out automatically. Compliance requires that all individual criteria pass.

Documentation Declaration Statements

1. The person who prepared the LMCV will sign and complete the fields for their name, company (if applicable), address, phone number, certification information (if applicable), date and signature.
2. The person who is assuming responsibility for the project being built to comply with Title 24, Part 6, will complete the fields (if applicable) for their company, responsible builder or installer name, CSLB license number, sample group number, dwelling test status in sample group, ECC Rater company name, ECC Rater name, ECC Rater signature, ECC Rater certification number and date signed.

A. Design ~~HERS~~ Verified Dwelling Unit Water Heating Systems Information (other than HPWH)
<<Static Text - see the top>><<If LMCC only has HPWH, display the "section does not apply" message>>
<<require one row of data for each water heater identified on the LMCC-PRF>>

01	02	03	04	05	06	07	08	09	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Central DHW System Distribution	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery
<<reference value from LMCC; >>	<<reference values from LMCC (see rule in header)>>	<<reference values from LMCC allowed values= DHW, Combined Hydronic, Hydronic>>	<<reference values from LMCC Allowed values = Boiler, Indirect, Consumer Instantaneous, Commercial Instantaneous, Consumer Storage, Commercial Storage, Residential-Duty Commercial Storage, or Residential-Duty Commercial Instantaneous >>	<<reference values from LMCC>>	<<reference values from LMCC. Allowed values are Heat Pump, Electric Resistance, Natural Gas, or Propane >>	<<If A06 = Heat Pump, then value = NA; Else reference values from LMCC. Allowed values: If Fuel Type A06 = Natural Gas, Propane then value = Btu/Hr. Else if Fuel Type = Electric Resistance then value = kW >>	<<if A06 = Heat Pump, then result = NA; If performance, reference values from LMCC-PRF; Elseif Prescriptive, then value = NA >	<<reference values from LMCC. If Central DHW System then value = Yes, else value = NA	<<reference values from LMCC If performance and A09 = NA then allowed values are *Standard Distribution System * Point of Use * Central Parallel Piping *Recirculation System Non-Demand Control * Demand Recirculation Manual Control * Demand Recirculation Sensor Control *~~HERS~~-Verified Pipe Insulation for Central Systems~~Credit~~ * ~~HERS~~-Verified Central Parallel Piping * ~~HERS~~-Verified Demand Recirculation Manual Control * ~~HERS~~-Verified Demand Recirculation Sensor Control; Else if A09 ≠ NA then allowed values are * Standard Distribution System *~~HERS~~-Verified Pipe Insulation for Central Systems~~Credit~~; Else if prescriptive, Allowed values are *Standard Distribution System *Demand Recirculation * Demand Recirculation Manual Control>>	<<reference values from LMCC. Allowed values are *Basic *Expanded *None>>	<<reference values from LMCC. Allowed values are *Yes *None>>

A2. Design ~~HERS~~ Verified Dwelling Unit HPWH System Information

<<Static Text - see the top>>

<<If LMCC has HPWH, then require table, else display the "section does not apply" message>>

<<require one row of data for each HPWH water heater identified on the LMCC-PRF>>

01	02	03	04	05	06	07	08	09	10
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Dwelling Unit DHW System Distribution Type	Compact Distribution	Drain Water Heat Recovery	Simulated Equipment Make and Model
<<reference value from LMCC >>	<<reference value from LMCC>>	<<references value from LMCC-PRF. Else if prescriptive, display "NEEA Tier 3" >>	<<references values from LMCC>>	<< Reference value from LMCC>>	<<reference values from LMCC-PRF; else NA>>	<<reference values from LMCC If performance, then allowed values are *Standard Distribution System * Point of Use * Central Parallel Piping *Recirculation System Non-Demand Control * Demand Recirculation Manual Control * Demand Recirculation Sensor Control *~~HERS-~~Verified Pipe Insulation for Central Systems~~Credit~~ * ~~HERS-~~ Verified Central Parallel Piping * ~~HERS-~~ Verified Demand Recirculation Manual Control * ~~HERS-~~ Verified Demand Recirculation Sensor Control; Else if prescriptive, Allowed values are *Standard Distribution System *Demand Recirculation * Demand Recirculation Manual Control>>	<<reference values from LMCC. Allowed values are *Basic *Expanded *None>>	<<reference values from LMCC. Allowed values are *Yes *None>>	<<if performance, hide column from user, needed for equivalency lookup; Reference value from XML; Elseif prescriptive, do not require field>>

B. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Systems Information

<<Static Text - see the top>><<If Table A does apply, then require table, else display the "section does not apply" message>>

<<require one row of data for each HPWH water heater identified in Table A>>

01	02	03	04	05	06	07	08	09	10	11	12
Dwelling Unit Name	Water Heating System ID or Name	Water Heating System Type	Water Heater Type	# of Like (or Identical) Water Heaters in System	Fuel Type	Rated Input Type	Rated Input Value	Central DHW System Distribution	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery
<<Reference value from A01 >>	<<Reference value from A02>>	<< reference value from A03>>	<< reference value from A04>>	<< reference value from A05>>	<< Reference value from A06>>	<< Reference value from A07 >>	<<User input value which must pass the following range tests: Value may be NA if B08is NA. If B06 = Natural Gas or Propane, then If B04 = Commercial Storage, then value must be > 75,000 Btu/hr; If B04 = Consumer Storage, then value must be ≤ 75,000 Btu/hr; If B04 = Commercial Instant, then value must be > 200,000 Btu/hr; If B04 = Consumer Instant, then value must be ≤ 200,000 Btu/hr; Else if B04 = Residential-Duty Commercial Storage, then value must be ≤ 105,000 Btu/hr;	<< Reference value from A09>>	<< Reference value from Design Dwelling Unit DHW System Distribution A10>>	<< Reference value from A11>>	<< Reference value from A12>>

							Else if B06 = Electricity, then If B04 = Commercial Storage or Commercial Instant, then value must be > 12 kW; If B04 = Consumer Storage or Consumer Instant, then value must be ≤ 12 kW; Else if B04 = Residential-Duty Commercial Instantaneous, then value must be ≤ 58.6 kW; If the value passes range test and if A06 = Electric Resistance, it is stored in WaterHeaterElectricFiredRatedInput, Otherwise the value is stored in WaterHeaterGasFiredRatedInput>>				

B2. Installed ~~HERS~~ Verified Dwelling Unit HPWH System Information

<<Static Text - see the top>><<If LMCCTable A2 does not apply, then require table, else display the "section does not apply" message>>

<<require one row of data for each water heater identified LMCCin table A2>>

01	02	03	04	05	06	07	08	09
Dwelling Unit Name	Water Heating System ID or Name	Modeled Equipment Make and Model	# of Like (or Identical) Water Heaters in System	Tank Location	Exterior Tank Insulation R-value	Dwelling Unit DHW System Distribution Type	Compact Distrib.	Drain Water Heat Recovery
<<Reference value from A2-01 >>	<<reference value from A2-02>>	<<if performance, user input is equal to A2-03 as default, and allow user to override with an equivalent system based on the simulated equipment in A2-10; elseif prescriptive newly constructed, If LMCI F16, option 2, then allow user to enter any 240V HPWH; Else if LMCI F16 option 3, then, allow user to enter any Tier 3 Or higher model>>	<<Reference value from A2-04>>	<<Reference value from A2-05.>>	<<User input value; check value must be ≥A2-06 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if A2-06 is NA >>	<< Reference values from A2-07>>	<< Reference values from A2-08>>	<< Reference values from A2-09>>
10	Compliance Statement:			<<calculated field: If A2-06 ≤ B2-06, then display result = System complies; else display result = system does not comply>>				

C. Design ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

<<Static Text - see the top>><< require one row of data for each water heater identified in Section A and A2 >>

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insulation R-Value	Water Heater Storage Volume (gal)	Tank location
<<Reference value from design Water Heating System ID or Name (A02) or A2-02>>	<< reference values from LMCC. If information is not available on LMCC, then allow user input.allowed values are: *Uniform Energy Factor *AFUE *Thermal Efficiency >>	<< reference values from LMCC-PRF-01; Else = NA>>	<<reference values from LMCC-PRF-01. Else = NA >>	<<reference values from LMCC-PRF-01. Else = NA >>	<<reference values from LMCC. NA is allowed only if Water Heater Type = Consumer Instantaneous, Commercial Instantaneous, or Residential-Duty Commercial Instantaneous >>	<<Reference value from LMCC>>

D. Installed ~~HERS~~ Verified Dwelling Unit Water Heating Efficiency Information

<<Static Text - see the top>><<require one row of data for each water heart identified in Section C >>

01	02	03	04	05	06	07
Water Heating System ID or Name	Heating Efficiency Type	Heating Efficiency Value	Standby Loss (%)	Exterior Insul. R-Value	Water Heater Storage Volume (gal)	Tank location
<<Reference value from design Water Heating System ID or Name (C01)>>	<<Reference value from Design Water Heater Type (C02). Value may be NA if C02 is NA >>	<<User input value; check value must be ≥ value in E03 to comply, else flag non-compliant values and do not allow the doc to be registered, . Value may be NA if C03 is NA >>	<<User input value; check value must be ≤ value in C04 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if C04 is NA >>	<<User input value; check value must be ≥ value in E05 to comply, else flag non-compliant values and do not allow the doc to be registered. Value may be NA if C05 is NA >>	<< User Input must equal reference values from E06, else flag non-compliant values and do not allow the doc to be registered. NA is allowed only if Water Heater Type = Consumer Instantaneous or Commercial Instantaneous >>	<<Reference value from C07 Value may be NA if LMCC value is NA >>
08	Compliance Statement	<<calculated field: If D03 = NA or D03 ≥ C03, D04 = NA or D04 ≤ C04, D05 = NA or D05 ≥ C05, D06 = C06, and D07 = C07, then display result = System complies; else display result = system does not comply>>				

E. Installed Water Heater Manufacturer Information <<require one row of data for each water heater identified in Section C >>		
01	02	03
Water Heating System ID or Name	Manufacturer	Model Number
<<reference value from B02 or B2-02>>	<<User input>>	<<User input>>

F. Mandatory Measures for all Domestic Hot Water Distribution Systems <<Static Text - see the top>>	
06	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.	

G. ~~HERS~~ Verified Compact Hot Water Distribution System Expanded Credit (CHWDS-H-EX) (RA3.6.5)

<<Static Text - see the top>><< Require one row of data, reporting the longest distances, for each dwelling identified in Table B or B2. with B11 or B2-08 = Expanded. If no dwelling in B11 or B2-08 = Expanded, then display section header and standard "This section does not apply" message>>

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor
<<Reference value from A01 or A2-01>>	<<if performance, then value = NA; Else if prescriptive, user select from list: 1, 2, 3>>	<<Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference value from LMCC-PRF; else if prescriptive or not in the XML and A10 or A2-07 = Standard Distribution System, then value = (G03*0.4) +(G04*0.4) +(G05*0.2); else if A10 or A2-07 = Demand Recirculation Manual Control, then value = G05>>	<< Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, value = ((a+b *CFA)/n) >> *Where: a, b = Qualification distance coefficients from Table 4.4.6-2 below, CFA = Conditioned floor area of the dwelling unit (ft^2) from LMCC, and n = Number of water heaters in the dwelling unit from A05 or A2-04 (unitless).*	<<Reference value from LMCC-PRF; elseif prescriptive, then value = n/a>>	<<Calculated value = 0.3 + 0.4 * (G06/G07), if performance, value must be ≤ G08; elseif prescriptive, value must be ≤ 0.7 >>

10	No hot water piping >1 inch diameter is allowed.	
11	Length of 1 inch diameter piping is limited to 8 feet or less.	
12	Two and three story buildings cannot have hot water distribution piping in the attic, unless the water heater is also located in the attic.	
13	Eligible recirculating systems must be ~~HERS~~-Verified Demand Recirculation: Manual Control conforming to RA4.4.17.	
14	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
15	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>	

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

H. Compact Hot Water Distribution ~~(CHWDS)~~ (RA4.4.6)

<<Static Text - see the top>><< Require one row of data, reporting the longest distances, for each dwelling identified in Section B or B2. with B11 or B2-08 = Basic. If no dwelling in B11 or B2-08 = Basic, then display section header and standard "This section does not apply" message>>

01	02	03	04	05	06	07	08	09
Dwelling Name	Number of Stories	Master Bath distance of furthest fixture to Water Heater in feet	Kitchen distance from furthest fixture to Water Heater in feet	Furthest Third furthest fixture to Water Heater in feet (Avg for multiple water heaters)	Weighted Distance	Qualification Distance	Design Compactness Factor	Calculated Compactness Factor
<<Reference value from A01 or A2-01>>	<<if performance, then value = NA; Else if prescriptive, user select from list: 1, 2, 3>>	<<Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, user input>>	<<Reference value from LMCC-PRF; else if prescriptive or not in the XML and A10 or A2-07 = Standard Distribution System, then value = (H03*0.4) +(H04*0.4) +(H05*0.2); else if A10 or A2-07 = Demand Recirculation Manual Control, then value = H05>>	<< Reference Value from LMCC-PRF; Else if prescriptive compliance or not in the XML, value = ((a+b *CFA)/n) >> *Where: a, b = Qualification distance coefficients from Table 4.4.6-2 below, CFA = Conditioned floor area of the dwelling unit (ft²) from LMCC, and n = Number of water heaters in the dwelling unit from A05 (unitless).*	<<Reference value from LMCC-PRF; elseif prescriptive, then value = n/a>>	<<Calculated value = 0.3 + 0.4 * (H06/H07), if performance, value must be ≤ H08; elseif prescriptive, value must be ≤ 0.7 >>

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met

Table 4.4.6-2: Coefficients for the Qualification Distance Calculation

<< do not show table, only use for equation in G07 and H07>>

	Coefficient a		**Coefficient b**	
Building Type	**Non-Recirculating**	**Recirculating**	**Non-Recirculating**	**Recirculating**
Multifamily (Non Central)	Use when distribution type (A10 Or A2-07) is *Standard Distribution System	Use when distribution type (A10 Or A2-07) is * Demand Recirculation Manual Control		
One story	7.5	n/a	0.0080	n/a

Two or more story	7.5	n/a	0.0050	n/a

I. ~~HERS~~ Verified Drain Water Heat Recovery System (DWHR-H) (RA3.6.9)

<<require one row of data for each drain water heat recovery system identified in Section B or B2. with B12 or B2-09 = Yes. Else report section header and standard “This section does not apply” message>>

Design DWHR System Information

01	02	03	04
System ID/Name	Rated Effectiveness	Installation Configuration	Percent of shower served by the DWHR device
<<Reference value from A02 or A2-02>>	<<If performance, reference value from LMCC-PRF. If prescriptive, display “≥ 42%”>>	<<If performance, reference value from LMCC-PRF. If prescriptive, display “NA”>>	<<If performance, reference value from LMCC-PRF If prescriptive, display “NA”>>

Installed DWHR System Information

05	06	07	08	09	10	11
System ID/Name	Manufacturer	Model #	Rated effectiveness	Installation Configuration	Percent of shower served by the DWHR device	DWHR System Certified by CEC
<<Reference value from B02 orB2-02>>	<<User input>>	<<User input>>	<<User input, value ≥ Rated Effectiveness in I02, 0.42≤I03<1 >>	<< If performance, reference value from I03. If prescriptive, user pick from list: • Equal flow • Unequal to shower • Unequal to water heater>>	<< If performance, reference value from I04. If prescriptive, User input,0<I06≤100 >>	□ Yes □ No

12	For water heating system serving a single dwelling, the DWHR system shall, at the minimum, recover heat from the master bathroom shower and must at least transfer that heat either back to the respective shower(s) or the water heater.	
13	For central water heating system serving multiple dwellings, the DWHR system shall, at the minimum, recover heat from half the showers located above the first floor and must at least transfer that heat either back to all the respective showers or the water heater.	
14	The DWHR unit(s) shall be installed within 1 degrees of the rated slope. Sloped DWHR shall have a minimum lengthwise slope of 1 degree. The lateral level tolerance shall be within plus or minus 1 degree.	
15	Verification Status:	☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
16	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>	

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

J. ~~HERS~~ Verified Pipe Insulation for Central Systems ~~Credit Requirements~~ (PIC-H) (RA3.6.~~3~~2) <<Static Text - see the top>><<If there are no systems in column A10 or A2-07 that have a value = "~~HERS~~-Verified Pipe Insulation for Central Systems~~Credit~~", then display the "section does not apply" message; else display this entire table >>		
01	~~HERS~~ECC rater shall perform a visual inspection that all hot water piping comply with the insulation requirements in 150.0(J).	
02	The rater shall use a sampling approach that one in seven DHW recirculation pipe risers and associated branches be inspected to verify the pipe insulation meet with the following requirements: a. All piping for multifamily domestic hot water systems shall be insulated to the thickness specified in Table 160.4-A, including the first 8 feet of inlet cold water piping to the heating plant. Insulation on the piping and appurtenances shall be continuous. b. All appurtenances at the heating plant, from a heating source to storage tank(s), or in between storage tanks and storage water heaters, and recirculation supply and return loop shall meet the following: 4. Insulation to be flush with pipe insulation or have minimum of one inch if appurtenance is bulkier. 5. Removable and re-installable for maintenance or replacement. 6. Pipe supports, hangers, and clamps shall be attached on the outside of rigid pipe insulation. c. All pipe insulation seams shall be sealed along the length of the pipe and between adjacent sections of insulation material. d. Insulation for pipe elbows shall be mitered, and insulation for tees shall be notched. Alternatively, tees and elbows may be pre-formed, or site fabricated with PVC covers. Isolation valves shall be fully functional. Extended stem isolation valves shall be installed on hot water piping or where pipe insulation is required.	
03~~2~~	Verification Status:	<<user pick from list: ☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
04~~3~~	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>	
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.		

K. ~~HERS~~ Verified Central Parallel Piping Requirements (PP-H) (RA3.6.4) <<Static Text - see the top>> <<If A10 or A2-07 = "~~HERS~~ECC-Verified Central Parallel Piping", then display this entire table, else display "section does not apply" message>>		
01	Each central manifold has 5 feet or less of pipe between manifold and water heater. (RA4.4.15)	
02	For manifolds that include valves, the manifold must be readily accessible in accordance with the plumbing code. (RA4.4.4)	
03	Hot water distribution system piping from the manifold to the fixtures and appliances must take the most direct path. For example, piping from a second story manifold cannot supply the first floor. (RA4.4.4)	
04	The hot water distribution piping must be separated by at least 2 inches from any other hot water supply piping, and at least 6 inches from any cold water supply piping. Alternatively, the hot water supply piping must be insulated to the thicknesses shown in TABLE 120.3-A. (RA 4.4.4)	
05	Verification Status:	<<user pick from list: ☐ Pass - all applicable requirements are met; or ☐ Fail - one or more applicable requirements are not met. Enter reason for failure in corrections notes field below; or ☐ All N/A - This entire table is not applicable
06	Correction Notes: <<if Verification Status= Fail, then text entry in this Corrections Notes field is required; user input text>>	
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.		

L. Central Parallel Piping Requirements (PP) (RA4.4.4) <<Static Text - see the top>> <<If A10 or A2-07 = "Central Parallel Piping", then display this entire table, else display "section does not apply" message>>
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

M. Point of Use Requirements (POU) (RA4.4.5)

<<Static Text - see the top>>

<<If A10 or A2-07 = “Point of Use”, then display this entire table, else display “section does not apply" message>>

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met.

N. Mandatory Requirements for all Recirculation System (RA4.4.7)

<<Static Text - see the top>> <<If A10 or A2-07 = “Recirculation System Non-Demand Control”, “Demand Recirculation Manual Control”, “Demand Recirculation Sensor Control”, “~~HERS~~ Verified Demand Recirculation Manual Control”, or “~~HERS~~ Verified Demand Recirculation Sensor Control”, then display this entire table, else display “section does not apply" message>>

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met

O. Recirculation Non-Demand Controls Requirements ~~(R-ND)~~ (RA4.4.8)

<<Static Text - see the top>> <<If A10 or A2-07 = “Recirculation System Non-Demand Control, then display this entire table, else display “section does not apply" message >>

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met.

P. Demand Recirculation Manual Control ~~(R-DRmc)~~ (RA4.4.9)/Sensor Control Requirements ~~(RDRsc)~~ (RA4.4.10)

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = “Demand Recirculation Manual Control”, “Demand Recirculation Sensor Control”, “~~HERS~~ Verified Demand Recirculation Manual Control”, or “~~HERS~~-Verified Demand Recirculation Sensor Control”, then display this entire table, else display “section does not apply" message >>

<<Static Text - see the top>>

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met.

Q. ~~HERS~~ Verified Demand Recirculation Manual Control (RDRmc-H) (RA3.6.6)/Sensor Control (RDRsc-H) (RA3.6.7) Requirements

Systems that utilize this distribution type shall comply with these requirements. <<If A10 or A2-07 = “~~HERS~~-Verified Demand Recirculation Manual Control” or “~~HERS~~-Verified Demand Recirculation Sensor Control”, then display this entire table, else display “section does not apply" message >>

<<Static Text - see the top>>

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met unless otherwise noted in the Verification Status and the Corrections Notes in this table.

R. Determination of ~~HERS~~ Verification Compliance

<<Static Text - see the top>>

01	<< if D08 result = System Complies, and results for all applicable sections F, G, I, J, K, and Q ≠ fail, then display: Complies: All specified verification protocol requirements on this document are met; else display: Does not comply: One or more specified verification protocol requirements on this document are not met.>>